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Evaluation of the microbiological quality of food products

CH5200 - Food Microbiology
Experiment 3



Group 2		
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1 General Introduction

Microorganisms, both beneficial and harmful, are commonly found in food products depending on various intrinsic factors (e.g., pH, water activity, nutritional content, and presence of antimicrobial compounds) and extrinsic factors (e.g., temperature, relative humidity, and composition of gases in the storage environment). Microbial contamination analysis is carried out to verify that food products remain within acceptable microbial limits, especially those susceptible to pathogenic contamination. This experiment aims to assess microbial load on a range of food products, namely fermented products, dried spices and herbs, and fresh products.

2 Materials and Methods

In this experiment, three food products were evaluated, namely five-spice powder (in a bottle), natto (styrofoam-packed), and enoki mushrooms (plastic-wrapped). All materials procured were store-bought, within their prescribed expiry dates, and visually evaluated to be without any serious defects or obvious spoilage. In particular, the natto and enoki mushrooms were kept in refrigeration throughout storage. Total viable counts (TVC) using PCA and fungal counts using DRBC were performed for the samples. Additional investigations for the presence of spore-forming bacteria, lactic acid bacteria, *C. perfringens*, and *Enterobacteriaceae spp.* were performed on relevant food products as well.

3 Results and Discussion

Five-spice powder, natto, and enoki mushrooms were processed using a stomacher. The produced homogenates were subjected to various microbial tests, which can be seen in Table 1.

Table 1: Recorded results of various microbial testing performed on five-spice powder, natto, and enoki mushrooms

Samples/ Tests	Five-spice powder	Natto	Enoki mushroom
Total Viable Count (TVC)	3.4	>8.8	8.4
Spore Forming bacteria (Spf)	2.8	4.8	Not performed
Yeast and Mold	<2.3	<2.3	5.3
<i>Clostridium perfringens</i>	<0.3	Not performed	Not performed
Lactic Acid Bacteria (LAB)	Not performed	<1.0	4.5
<i>Enterobacteriaceae</i>	Not performed	Not performed	<3.0

3.1. Herbs and spices (Five-spice powder)

Five-spice powder is a common seasoning used in Chinese cuisine and is usually composed of a blend of star anise, cloves, Chinese cinnamon, Sichuan peppercorns, and fennel seeds, in a dried powdered form. From Table 1, five-spice powder displayed a high sterility for all tests conducted, being under the lower threshold for detection for the spore former, yeasts, molds and *C. perfringens* tests. The TVC and spore former count (3.4 and 2.8 log₁₀CFU/g) were both moderately high, suggesting that the microflora of five-spice powder was dominated by spore forming bacteria, which was expected considering the low water activity of dried spices.

In general, food products with microbial counts below 6.0 log₁₀CFU/g are recommended for safe consumption; counts between 6.0 and 7.0 log₁₀CFU/g are considered acceptable, while those exceeding 7.0 log₁₀CFU/g are deemed unsatisfactory and potentially unsafe (Kokkinakis et al., 2020). A TVC result of 3.4 could therefore be regarded as being unquestionably within the safe ranges. Meanwhile, a safety limit of 4.0 log₁₀CFU/g is recommended for sporeformers, yeast, and molds (Berg et al., 2017).

Five-spice powder was once again found to be within acceptable safety limits across all three aforementioned tests. Furthermore, the test for the common bacterium *Clostridium perfringens*, which is commonly found in many herbs and spices, produced a low reading of <0.3 MPN/g, indicating its safety for consumption.

Compared to the results from other spices (Annex Table B1), five-spice had a relatively low microbial count, suggesting that its microbiological quality was on par with black peppercorns, star anise and Italian herbs. This may be attributed to the antimicrobial activity of cloves and cinnamon, which were present in the five-spice mix. Cloves and cinnamon contain eugenol and cinnamaldehyde respectively, which can act as antimicrobials that inhibit the bacterial and fungal growth (Adams et al., 2024).

3.2. Fermented foods (Natto)

Natto demonstrated significantly low and safe levels in yeast, mold, and lactic acid bacteria (LAB) tests. However, it showed relatively high counts in both the TVC and spore former tests. This was expected because natto is intentionally inoculated with a spore forming bacterium—*Bacillus subtilis* to act as a fermentative agent. Most typically, fermented foods are intentionally inoculated with microbes, and thus a high count is not necessarily indicative of spoilage. Thus, the standard upper threshold for microbial spoilage (7.0 log₁₀CFU/g) is not always applicable for fermented products.

Compared to the results from other fermented products (Annex Table B2), natto stands out among the tested fermented food products with the highest TVC reading at >8.8 log₁₀CFU/g, significantly surpassing other samples such as kimchi (7.0 log₁₀CFU/g), tempeh (7.6 log₁₀CFU/g), and yogurt (3.8 log₁₀CFU/g), indicating a highly dense microbial population. Additionally, natto showed a relatively high spore former count at 4.8 log₁₀CFU/g, further supporting the presence of *Bacillus subtilis* and its spore-forming capability. Conversely, other samples, including kombucha, Yakult, and miso paste, displayed significantly lower TVC and spore former readings, suggesting that natto's distinct microbial profile is characterized by both its robust fermentative bacterial activity and spore formation.

3.3. Fresh produce (*Enoki mushroom*)

Enoki mushrooms were subjected to TVC, yeast and mold, LAB, and *Enterobacteriaceae* testing. Enoki yielded a notably high TVC of 8.4 log₁₀CFU/g, which was above the typical spoilage threshold level of 7.0 log₁₀CFU/g. This could be attributed to the fact that the enoki samples used for testing included the entire fungal structure—encompassing the stipe, base, hyphae, and mycelium at the lower portion. The bottom section may retain soil or dirt residues and is typically trimmed and discarded prior to consumption, which could have contributed to the higher microbial readings observed.

Detected microbes include LAB, spore forming bacteria, yeasts and molds, all present at relatively high levels (around 4.5 to 5.3 log₁₀CFU/g). Notably, the *Enterobacteriaceae* test yielded results that were below the detection limit, unexpectedly suggesting their absence. This could be due to *Enterobacteriaceae* strains being outcompeted by other more dominant strains, such as *L. monocytogenes*—a common cause of listeriosis outbreaks originating from enoki mushrooms (Pereira et al., 2023). Additionally, the presence of LAB may have created a low pH environment or bacteriocins such as nisin, that may inhibit the growth of *Enterobacteriaceae*.

Compared to the results from other vegetable and edible fungi products (Annex Table B3), enoki mushrooms and dou miao (pea sprouts) were compared as both are vegetable-based products, making them suitable for assessing differences in microbial profiles. Dou miao showed a TVC reading of more than 9.26 log₁₀CFU/g, whereas enoki mushrooms had a TVC reading of 8.4 log₁₀CFU/g, indicating a significantly higher microbial load. In contrast to enoki mushrooms—which had lower levels of both mold and yeast (5.3 log₁₀CFU/g)—dou miao also displayed noticeably higher levels of both (8.0 log₁₀CFU/g). Additionally, LAB were more prevalent in enoki mushrooms (5.8 log₁₀CFU/g) than in dou miao (<3.0 log₁₀CFU/g). In contrast, *Enterobacteriaceae* was detected at >7.5 log₁₀CFU/g in dou miao but remained <1.0 log₁₀CFU/g in enoki mushrooms. These noticeable differences may result from dou miao's higher moisture content, delicate tissue structure, and higher potential for spoilage, all of which foster microbial growth. In addition, compared to the relatively better-preserved enoki mushrooms, dou miao showed visible signs of spoilage by the time of microbial investigation, which was potentially caused by poor handling, storage conditions, and time since harvest.

4 Conclusions and Recommendations

In conclusion, five-spice powder exhibited outstanding microbial quality, which was most likely attributed to its low water activity and natural antimicrobial properties from various compounds, such as eugenol and cinnamaldehyde. Although natto displayed high TVC and sporeformer counts, it remained safe for consumption as natto production relies on the intentional inoculation of *Bacillus subtilis*—the main fermentative bacteria for natto. Conversely, enoki mushrooms displayed elevated microbial loads, especially in TVC testing, raising concerns over spoilage and potential contamination. It is generally recommended that enoki mushrooms be thoroughly trimmed, washed, and cooked before consumption. Storage conditions could also be optimized to reduce microbial proliferation. Moving forward, greater attention should be placed on promoting public education regarding appropriate food preparation techniques. Ongoing microbiological monitoring is recommended, particularly for high-risk products, to ensure greater food safety.

References

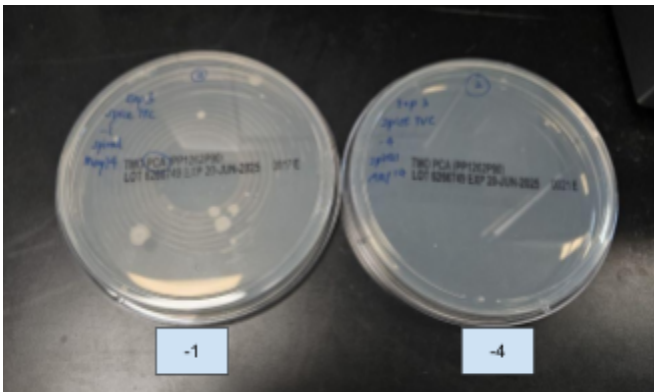
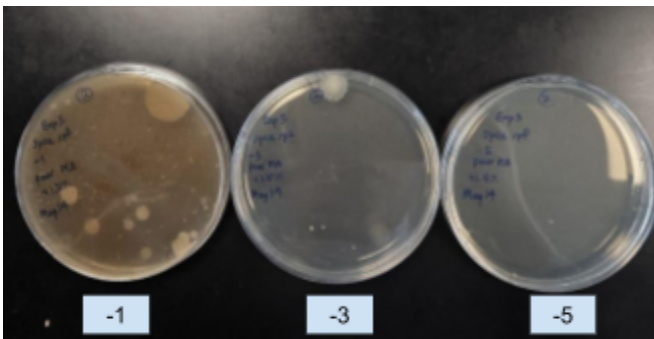
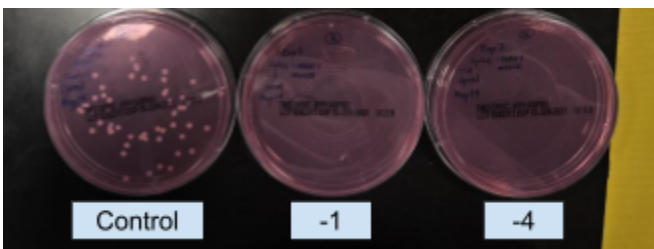
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Annex

Table A1: Microbiological quality of selected herbs and spices—five-spice powder

Group of m.o.	Medium	Type of plate	Lower detection limit (log ₁₀ CFU /g)	Dilution factor	CFU counted (Prelimin ary)	CFU counted (After confirmati on)	Concentration/ log ₁₀ CFU/g
TVC	PCA	Spiral plate	2.3	-1	12	12	12 x 10 x 20 = 2400 log ₁₀ (2.4 x 10 ³) = <u>3.4</u>
				-4	<1	<1	
Spf	PCA	Pour plate	1.0	-1	57	57	57 x 10 x 1 = 570 log ₁₀ (5.7 x 10 ²) = <u>2.8</u>
				-3	7	7	
				-5	<1	<1	
Yeast	DRBC	Spiral plate	2.3	Control	52	54	1 x 10 x 20 = 200 log ₁₀ (2 x 10 ²) = <u><2.3</u>
				-1	<1	<1	
				-4	<1	<1	
C. perfrin gens MPN/g	N/A				<1	<1	MPN code: 0-0-0 <u><0.3</u>

Table A2: Images of the tests used to determine the microbiological quality of selected herbs and spices—five-spice powder

Group of m.o.	Dilution factor	CFU counted (Preliminary)	CFU counted (After confirmation)	Results
TVC	-1	12	12	
	-4	<1	<1	
Spf	-1	57	57	 *Sample for -1 dilution is slightly brown not due to microbial growth, but the colour of the five-spice suspension
	-3	7	7	
	-5	<1	<1	
Yeast	Control	52	54	
	-1	<1	<1	
	-4	<1	<1	

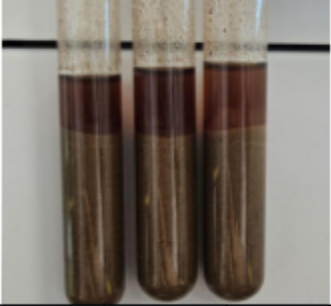
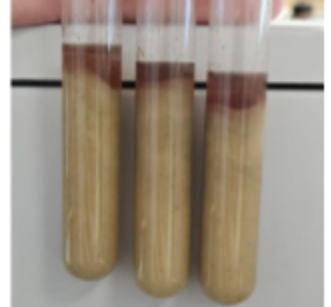
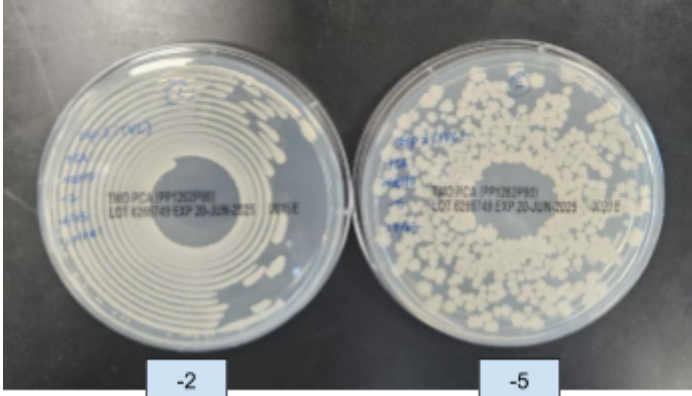
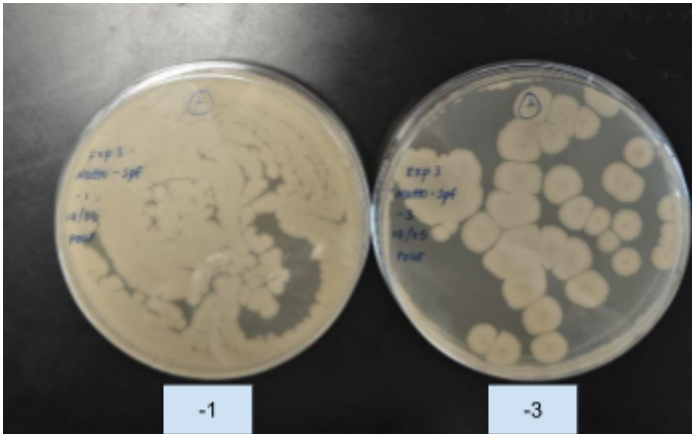
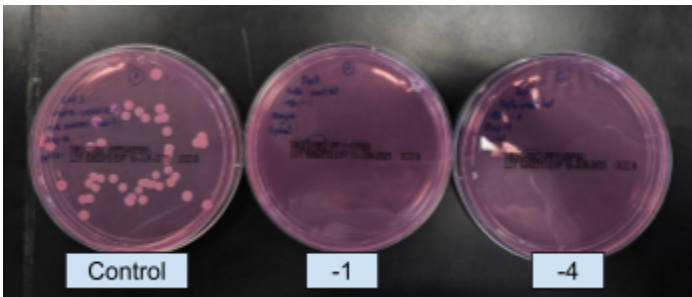
C. perfrin gens MPN/g	-1a, -1b, -1c, -2a, -2b, -2c, -3a, -3b, -3c	<1	<1	 -1a -1b -1c  -2a -2b -2c  -3a -3b -3c *All samples showed no stormy fermentation, no gas formation or coagulation (brown sediments at the bottom were found to be the five-spice powder)
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Table A3: Microbiological quality of selected fermented product—natto

Group of m.o.	Medium	Type of plate	Lower detection limit ($\log_{10}$ CFU/g)	Dilution factor	CFU counted (Preliminary)	CFU counted (After confirmation)	Concentration/ $\log_{10}$ CFU/g
TVC	PCA	Spiral plate	2.3	-2	>300	>300	$300 \times 10^5 \times 20 = 600000000$ $\log_{10}(6 \times 10^8) = \underline{\underline{8.8}}$
				-5	>300	>300	
Spf	PCA	Pour plate	1.0	-1	>300 (overgrown)	>300 (overgrown)	$61 \times 1000 \times 1 = 61000$ $\log_{10}(6.1 \times 10^4) = \underline{\underline{4.8}}$
				-3	61	61	
Yeast	DRBC	Spiral plate	2.3	Control	46	50	$1 \times 10^1 \times 20 = 200$ $\log_{10}(2 \times 10^2) = \underline{\underline{2.3}}$
				-1	<1	<1	
				-4	<1	<1	
LAB	MRSA	Pour plate	1.0	Control	>300 (overgrown)	>300 (overgrown)	$1 \times 10 = 10$ $\log_{10}(10) = \underline{\underline{1.0}}$
				-1	<1	<1	
				-3	<1	<1	
				-5	<1	<1	

Table A4: Images of the tests used to determine the microbiological quality of selected fermented product—natto

Group of m.o.	Dilution factor	CFU counted (Preliminary)	CFU counted (After confirmation)	Results
TVC	-2	>300	>300	
	-5	>300	>300	
Spf	-1	>300 (overgrown)	>300 (overgrown)	
	-3	61	61	
Yeast	Control	46	50	
	-1	<1	<1	
	-4	<1	<1	

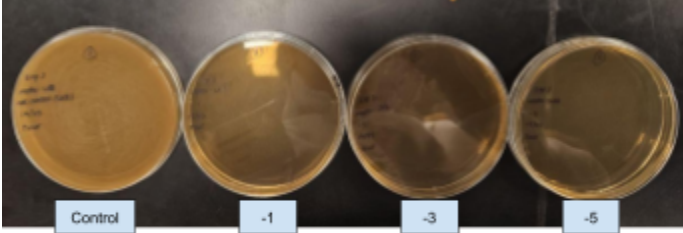
LAB	Control	>300 (overgrown)	>300 (overgrown)	
	-1	<1	<1	
	-3	<1	<1	
	-5	<1	<1	

Table A5: Microbiological quality of selected edible fungi—enoki mushroom

Group of m.o.	Medium	Type of plate	Lower detection limit ($\log_{10}$ CFU/g)	Dilution factor	CFU counted (Preliminary)	CFU counted (After confirmation)	Concentration/ $\log_{10}$ CFU/g
TVC	PCA	Spiral plate	2.3	-2	>300	>300	$115 \times 10^5 \times 20 = 230,000,000$ $\log_{10}(2.3 \times 10^8) = \underline{\underline{8.4}}$
				-5	115	115	
Yeast	DRBC	Spiral plate	2.3	Control	44	44	$1 \times 10^4 \times 20 = 200000$ $\log_{10}(2 \times 10^5) = \underline{\underline{5.3}}$
				-1	>300	>300	
				-4	1	1	
LAB	MRSA	Pour plate	1.0	Control	>300	>300	$35 \times 10^3 = 35000$ $\log_{10}(3.5 \times 10^4) = \underline{\underline{4.5}}$
				-3	61	$61 \times (4/7) = 35$	
				-5	<1	<1	
<i>Enterobacteriaceae</i>	VRBG	Pour plate	1.0	Control	>300	>300	$1 \times 10^3 = 1000$ $\log_{10}(10^3) = \underline{\underline{3.0}}$
				-3	<1	<1	
				-5	<1	<1	

Table A6: Images of the tests used to determine the microbiological quality of selected edible fungi—enoki mushroom

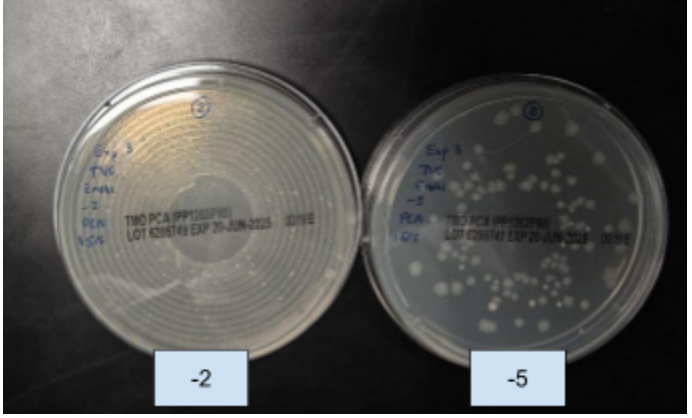
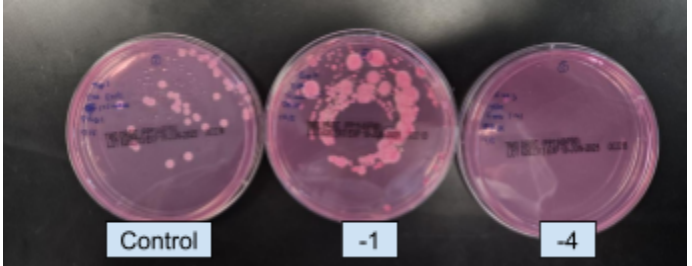
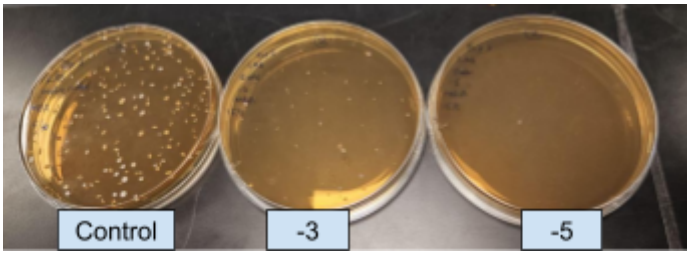
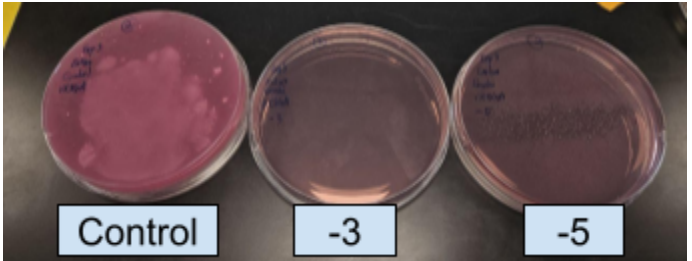
Group of m.o.	Dilution factor	CFU counted (Preliminary)	CFU counted (After confirmation)	Results
TVC	-2	>300	>300	
	-5	115	115	
Yeast	Control	44	44	
	-1	>300	>300	
	-4	1	1	
LAB	Control	>300	>300	
	-3	61	61 x (4/7) = 35	
	-5	<1	<1	
<i>Enterobacteriaceae</i>	Control	>300	>300	
	-3	<1	<1	
	-5	<1	<1	

Table B1: Comparison of microbiological quality of selected herbs and spices across all groups

Category 1: Herbs and Spices					
Product	Total Viable Count (TVC)/ log ₁₀ CFU/g	Spore-forming bacteria (spf), spore concentration/ log ₁₀ CFU/g	Yeast concentration/ log ₁₀ CFU/g	Mold concentration/ log ₁₀ CFU/g	<i>C. perfringens</i> concentration, Most Probable Number (MPN)/ MPN/g
Turmeric Powder	6.5	6.7	2.8	<2.3	2.9
Five Spice Powder	3.4	2.8	<2.3	<2.3	<0.3
Rosemary	4.3	2.4	<2.3	<2.3	<0.3
Cinnamon Powder	4.0	4.5	<2.3	<2.3	0.36
Jasmine Tea	3.3	2.6	<2.3	3.3	4.3
Italian Herbs	<2.3	2.6	<2.3	<2.3	0.92
Black Peppercorn	<2.3	<1.0	<2.3	<2.3	<0.3
Star Anise	<2.3	<1.0	<2.3	<2.3	<0.3
Dried Chrysanthemum Flower	4.6	4.4	<2.3	<2.3	0.92
Thyme	3.8	5.1	<2.3	<2.3	<0.3

Table B2: Comparison of microbiological quality of selected fermented products across all groups

Category 2: Fermented Products					
Product	Total Viable Count (TVC)/ log ₁₀ CFU/g	Spore-forming bacteria (spf), spore concentration/ log ₁₀ CFU/g	Yeast concentration/ log ₁₀ CFU/g	Mold concentration / log ₁₀ CFU/g	Lactic Acid Bacteria (LAB) concentration/ log ₁₀ CFU/g
Kombucha	<2.3	<2.3	2.8	<2.3	<2.0
Natto	>8.8	4.8	<2.3	<2.3	<1.0
Kimchi	7.0	2.8	3.7	3.7	>7.5
Yogurt	3.8	5.2	5.3	5.3	2.6
Tempeh	7.6	3.3	6.9	7.2	>7.5
Yakult	<2.3	<2.3	<2.3	<2.3	>7.5
Fermented Soybean Paste	5.7	>3.5	<2.3	<2.3	<1.0
Honey Mead	3.6	<2.3	2.4	<2.3	<2.3
Miso Paste	3.3	2.8	<2.3	<2.3	1.5
Belacan (Fermented Shrimp Paste)	3.8	3.1	<2.3	<2.3	<1.0

Table B3: Comparison of microbiological quality of selected fungi, vegetables and fruits across all groups

Category 3A: Fungi					
Product	TVC/ log ₁₀ CFU/g	Yeast concentration/ log ₁₀ CFU/g	Mold concentration/ log ₁₀ CFU/g	LAB concentration/ log ₁₀ CFU/g	<i>Enterobacteriaceae</i> concentration/ log ₁₀ CFU/g
Enoki mushroom	8.4	5.3	5.3	5.8	<1.0
Category 3B: Vegetables					
Product	TVC/ log ₁₀ CFU/g	Yeast concentration/ log ₁₀ CFU/g	Mold concentration/ log ₁₀ CFU/g	LAB concentration/ log ₁₀ CFU/g	<i>Enterobacteriaceae</i> concentration/ log ₁₀ CFU/g
Dou miao (Pea sprouts, noticeably spoilt by the time of investigation)	>9.3	8.2	8.2	<3.0	>7.5
Category 3C: Fruits					
Product	TVC/ log ₁₀ CFU/g	Yeast concentration/ log ₁₀ CFU/g	Mold concentration/ log ₁₀ CFU/g	LAB concentration/ log ₁₀ CFU/g	<i>Enterobacteriaceae</i> concentration/ log ₁₀ CFU/g
Fruit cup (Pre-cut fruit from one day before the time of investigation)	5.4	5.3	5.3	4.9	>3.5

Table B4: Comparison of microbiological quality of selected raw meat/ fish, ice cream and dairy, bread or pastry products across all groups

Category 4A: Raw Meat/ Fish								
Product	TVC/ log ₁₀ CFU/ g	Spf, spore concentra tion/ log ₁₀ CFU/ g	Yeast concentr ation/ log ₁₀ CFU/ g	Mold concentr ation/ log ₁₀ CFU /g	LAB concentr ation/ log ₁₀ CFU /g	<i>Enterobacteriac</i> <i>eae</i> concentration/ log ₁₀ CFU/g	<i>Pseudo</i> <i>monas</i> concentr ation/ log ₁₀ CFU /g	<i>Bacillus</i> <i>Cereus</i> concentr ation/ log ₁₀ CFU /g
Raw Minced Chicken	>7.8	Not done	4.1	Not done	Not done	2.6	>4.8	Not done
Ready-To-Eat Shrimp	2.8	Not done	Not done	Not done	<1.0	<1.0	3.3	Not done
Raw Squid	4.7	Not done	Not done	Not done	<1.0	<1.0	<2.3	Not done
Raw Salmon (Non-sashimi grade)	6.5	Not done	Not done	Not done	<1.0	>3.5	5.6	Not done
Category 4B: Water, milk, ice cream or dairy								
Product	TVC/ log ₁₀ CFU/ g	Spf, spore concentra tion/ log ₁₀ CFU/ g	Yeast concentr ation/ log ₁₀ CFU/ g	Mold concentr ation/ log ₁₀ CFU /g	LAB concentr ation/ log ₁₀ CFU /g	<i>Enterobacteriac</i> <i>eae</i> concentration/ log ₁₀ CFU/g	<i>Pseudo</i> <i>monas</i> concentr ation/ log ₁₀ CFU /g	<i>Bacillus</i> <i>Cereus</i> concentr ation/ log ₁₀ CFU /g
Pastry cream	4.8	<1.0	Not done	Not done	<1.0	3.5	<1.0	<1.0
Chocolate sundae	2.8	<1.0	Not done	Not done	<1.0	<1.0	<2.3	<1.0
Category 4C: Bread or pastry								
Product	TVC/ log ₁₀ CFU/ g	Spf, spore concentra tion/ log ₁₀ CFU/ g	Yeast concentr ation/ log ₁₀ CFU/ g	Mold concentr ation/ log ₁₀ CFU /g	LAB concentr ation/ log ₁₀ CFU /g	<i>Enterobacteriac</i> <i>eae</i> concentration/ log ₁₀ CFU/g	<i>Pseudo</i> <i>monas</i> concentr ation/ log ₁₀ CFU /g	<i>Bacillus</i> <i>Cereus</i> concentr ation/ log ₁₀ CFU /g
Kimbap (Rice roll)	8.9	<1.0	7.4	<2.3	>6.4	<1.0	Not done	<2.0